Topic: Speech Recognition

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Dave Touretzky edited this page on 18 Oct 2018 · 13 revisions

These are working notes for a topic area

Grade Progression: What should students know?

- **Grades K-2:** Some types of devices can recognize human speech. This includes most cellphones, and home entertainment systems like Amazon's Echo or Google Home.
- **Grades 3-5:** Speech recognition systems use grammatical knowledge to disambiguate homophones such as bear/bare or there/their/they're. Example: "*There* is no hot water" vs. "*Their* hot water is off" vs. "*They're* waiting for the hot water to come back on".
- **Grades 6-8:**
- **Grades 9-12:**

Readings for Working Group

1. Machine Learning is Fun Part 6: How to do Speech Recognition with Deep Learning. Adam Geitgey, Medium, December 2016. medium.com
2. How Speech Recognition Works. Sudeesh Puthiyedath, CodeGuru.com, August 3, 2006. part 1 and part 2
3. How Siri Works -- Interview with Tom Gruber, CTO of SIRI. Nova Spivack, NovaSpivack.com, January 26, 2010. novaspivack.com

> ### Old Readings (replaced)
>
> a. Brief Explanation of AI for Layman medium.com
>
> b. Making the Leap from Speech to Dialogue: The Challenge for Human to Machine Communication medium.com
>
> c. CACM January 2014 - A historical perspective of speech recognition. Xuedong Huang, James Baker, and Raj Reddy. Commun. ACM 57, 1 (January 2014), 94-103. DOI: acm.org

> d. CACM April 2018 - Speech emotion recognition: two decades in a nutshell, benchmarks, and ongoing trends. Björn W. Schuller. Commun. ACM 61, 5 (April 2018), 90-99. DOI: doi.org/10.1145/3129340 acm.org
>
> e. Video: Speech Emotion Recognition. youtube.com

Demo Resources

- Demo: Speech Recognition in Chrome
- Alexa, Siri, Cortana
- Use Audacity (free download) to record a speech segment and display the spectrogram.

Miscellaneous concepts to incorporate

Audio -> Formants -> Phones -> Syllables -> Words -> Phrases

How neural nets improved speech recognition: use of massive training data.

Grammar: recognition does best with conversational English

"How to recognize speech" == "How to wreck a nice beach"

Languages other than English

Accents; child voices

Applications: Alexa, Siri, Cortana. What do they do? How are they useful?

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2018 Fall Symposium Materials

Book Series: The World of Artificial Intelligence 4D

Book: Artificial Intelligence: A Modern Approach

Book: What to Think About Machines That Think

Competition: Halite

Competition: WAICY

Demo: Speech Recognition in Chrome

Demo: TensorFlow Playground

Documentary: AlphaGo

Knowledge: What Is Artificial Intelligence?
Online Course: Artificial Intelligence Explorations and Their Practical Use in Schools
Resource Directory
Resource: AIinSchools
Software: Calypso for Cozmo
Software: Cognimates
Software: eCraft2Learn
Software: Machine Learning For Kids
Topic: Machine Learning
Topic: Speech Recognition
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Working and Advisory Group Members

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```
https://github.com/touretzkyds/ai4k12.wiki.git
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